PRACTICAL-1

Identify Internal Components of a Desktop Computer and Observe Their Functions

A. Objective

To identify internal components of a desktop computer and observe their functions.

B. Expected Program Outcomes (POs)

PO1: Apply knowledge of engineering fundamentals to identify the internal components of a desktop computer and explain their functions.

PO5: Apply appropriate hardware resources and engineering tools to observe and identify internal computer components during laboratory practice.

PO8: Function effectively as an individual or as a member of a team while identifying and observing internal computer components.

PO9: Communicate observations by documenting the identified hardware components and their functions in a clear and systematic manner.

C. Expected Skills to be Developed Based on Competency

1. Identify internal components.
2. Explain their functions.
3. Observe hardware interconnections.

D. Expected Course Outcomes (COs)

Identify and elaborate components of a PC/laptop and overall system architecture.

E. Practical Outcome (PrO)

1. Identify internal hardware components.
2. Explain the function of each component.

F. Expected Affective Domain Outcomes (ADOs)

1. Follow laboratory safety.
2. Work effectively in a team.

G. Prerequisite Theory

A desktop computer is an electronic device that accepts data as input, processes it according to a set of instructions, stores it, and produces meaningful information as output. The internal components of a desktop computer work together to perform these operations efficiently.

The system unit (computer cabinet) houses the major internal hardware components such as the motherboard, processor, memory, storage devices, and power supply. Understanding these components and their functions is essential for computer assembly, maintenance, troubleshooting, and upgrading.

Major Internal Components of a Desktop Computer

1. Motherboard

The motherboard is the main printed circuit board (PCB) of the computer. It connects all hardware components and enables communication between them through buses and chipsets.

Functions**:**

- Holds and connects all internal components.
- Provides CPU and RAM sockets.
- Contains BIOS/UEFI firmware.
- Distributes power to connected devices.
- Supports expansion cards and peripheral interfaces.

2. Central Processing Unit (CPU)

The CPU is the brain of the computer. It performs arithmetic calculations, logical operations, and controls the execution of all instructions.

Functions:

- Executes program instructions.
- Performs arithmetic and logical operations.
- Controls data processing and system operations.

3. CPU Heat Sink and Cooling Fan

The CPU generates heat during operation. The heat sink absorbs this heat, while the cooling fan dissipates it to maintain a safe operating temperature.

Functions:

- Prevents CPU overheating.
- Improves system stability and performance.
- Increases processor life.

4. Random Access Memory (RAM)

RAM is the primary (volatile) memory used to store programs and data currently being processed by the CPU.

Functions**:**

- Stores active programs and data.
- Provides high-speed access to the processor.
- Loses data when power is switched OFF.

5. Storage Devices (HDD/SSD)

Storage devices provide permanent (non-volatile) storage for the operating system, applications, and user files.

Types:

- Hard Disk Drive (HDD): Magnetic storage with high capacity.
- Solid State Drive (SSD): Flash-based storage with faster speed and higher reliability.

6. Switched Mode Power Supply (SMPS)

The SMPS converts AC mains power into regulated DC voltages required by the computer.

Functions:

- Supplies +3.3 V, +5 V, and +12 V DC.
- Protects components from voltage fluctuations.
- Distributes power to all internal devices.

7. Expansion Cards

Expansion cards are installed in motherboard expansion slots to provide additional functionality.

Examples**:**

- Graphics Card (GPU)
- Sound Card
- Network Interface Card (NIC)
- Wi-Fi Adapter

8. CMOS Battery

The CMOS battery supplies power to the Real-Time Clock (RTC) and preserves BIOS settings when the computer is switched OFF.

Interconnection of Components

All internal components communicate through the motherboard using:

- Data Bus
- Address Bus
- Control Bus

The CPU exchanges data with RAM, storage devices, and input/output controllers through these buses.

H. Work Situation

Faculty demonstrates the internal components of a desktop computer. Students identify each component and observe its function.

I. Resources Required

Desktop computer, screwdriver, anti-static wrist strap.

J. Procedure

1. Switch OFF the computer and disconnect the power cable from the power supply.
2. Remove the side panel of the cabinet using a screwdriver.
3. Identify the major internal components such as the Motherboard, CPU, CPU Fan, RAM, SMPS, HDD/SSD, CMOS Battery, Expansion Cards, and SATA cables.
4. Observe the location and function of each component without disturbing their connections.
5. Trace the power and data cable connections between the SMPS, motherboard, and storage devices.
6. Record the name, location, and function of each identified component in the observation sheet.
7. Refit the side panel, reconnect the power cable, and switch ON the computer.
8. Verify that the system boots normally.
9. Clean the work area and submit the completed observation sheet.

K. Practical Related Quiz

1. Define Hardware and Software.

2. List Different parts of Desktop Computer Shown in fig 1



Fig. 1 Desktop Computer

3. List Different parts of Laptop System shown in fig.



Fig.2 Laptop

(Space for Answers)